

A nonempty counterexample to conjecture 00000009700

ColdSmile October 5, 2026

Abstract

The conjecture asserts that every algebraically independent set of transcendental numbers has Hausdorff dimension one. A singleton consisting of a transcendental real number is algebraically independent over the rationals and has Hausdorff dimension zero. This refutes an explicit necessary clause of the conjecture.

1 Source and conventions

The English source begins its conjecture with:

Every algebraically independent set of transcendental numbers has full Hausdorff dimension 1, and the box dimension spectrum of its complement is an explicit subinterval of $[0, 1/2]$; algebraically independent number sets are full-measure dense G-delta under Lebesgue measure.

The Chinese version likewise universally quantifies over algebraically independent sets of transcendental numbers and requires Hausdorff dimension one. Neither version imposes infinitude, maximality, or a transcendence-basis condition. The complete bilingual source is included as `conjecture.md`.

We use real numbers with their ordinary metric, and algebraic independence and transcendence over $\mathbb{Q}$. These are explicit conventions: the source does not name the ground field or ambient space. The real-line interpretation is supported by its description of dimension one as full and its use of Lebesgue measure.

For $S \subseteq \mathbb{R}$, algebraic independence means that evaluation on its inclusion $S \hookrightarrow \mathbb{R}$ is injective on the polynomial algebra $\mathbb{Q}[X_s : s \in S]$. A member t is transcendental if no nonzero polynomial in $\mathbb{Q}[X]$ vanishes at t . Hausdorff dimension is the standard metric dimension obtained from Hausdorff measures. Write

$$H(S) :\iff S \text{ is algebraically independent over } \mathbb{Q} \quad \text{and} \quad (\forall t \in S) \, t \text{ is transcendental over } \mathbb{Q}.$$

The precise first clause to be refuted is

$$(\forall S \subseteq \mathbb{R}) \quad H(S) \implies \dim_{\mathrm{H}} S = 1. \tag{1}$$

2 Counterexample and proof

Lemma 1. *There exists a transcendental real number.*

Proof. There are countably many polynomials with rational coefficients, and each nonzero such polynomial has finitely many real roots. The algebraic real numbers therefore form a countable union of finite sets and are countable. Since $\mathbb{R}$ is uncountable, some real number is not algebraic over $\mathbb{Q}$. $\square$

Theorem 1. *There exists a nonempty set $S \subseteq \mathbb{R}$ for which $H(S)$ holds and $\dim_{\mathbb{H}} S = 0$. Consequently, (1) is false.*

Proof. Choose a transcendental real number t and let $S = \{t\}$. This set is nonempty and every member is transcendental. The polynomial algebra with its single variable indexed by S is the ordinary one-variable polynomial algebra over $\mathbb{Q}$. Evaluation at t has zero kernel by transcendence, so it is injective. Thus the actual inclusion of S into $\mathbb{R}$ is algebraically independent.

For every exponent $s > 0$, a singleton can be covered by an interval with arbitrarily small positive diameter. The s th power of that diameter tends to zero, so its s -dimensional Hausdorff measure is zero. It follows that its Hausdorff dimension is zero. Applying (1) to this S would give $0 = 1$, a contradiction. $\square$

3 Relationship to the complete conjecture

The asserted dimension equality is a necessary conjunct in both source versions. The remaining assertions occur as conclusions, not as extra hypotheses restricting S . Failure of (1) therefore refutes the stated conjunction. The phrase “box dimension spectrum” has no parameter or convention specified in the source; this disproof does not assign it a new definition. The witness is finite and nonempty, as permitted by both source versions.

4 Formal verification

The accompanying Lean project uses the pinned Lean 4.19.0 and Mathlib v4.19.0. It formalizes real sets, the inclusion-indexed `AlgebraicIndependent` predicate over $\mathbb{Q}$, `Transcendental`, and the genuine Hausdorff dimension `dimH`. The dimension values are extended nonnegative reals. All new declarations are in namespace `Conjecture9700`. The theorem `exists_nonempty_counterexample` gives the actual nonempty witness, both source hypotheses, dimension zero, and failure of dimension one. The main theorem `hausdorff_clause_false` proves $\neg \text{HausdorffClause}$. The theorem `universal_conjunction_false` establishes the same obstruction for every additional predicate on the qualifying sets; it assigns no mathematical meaning to the unspecified spectrum.

A fresh independent build and warning-as-error replays succeeded. All ten compiled declarations (one definition and nine theorems) were inventoried and inspected; their dependencies use only `propext`, `Classical.choice`, and `Quot.sound`. The complete type-and-body dependency audit found no unsafe, partial, or missing dependencies. The source contains no admitted proofs, `native_decide`, or added axioms. Execution logs, theorem types, source identities, and reproduction instructions accompany the submission. No numerical approximation or auxiliary numerical computation is needed for the mathematics.

References

- [1] The Stacks Project, Definition 9.26.1, “Transcendence,” tag 030E.
- [2] Mathlib v4.19.0, modules `RingTheory.AlgebraicIndependent.Defs`, `RingTheory.Algebraic.Defs`, and `Topology.MetricSpace.HausdorffDimension`.